

Schizophrenia Classification from EEG Signals

Classical Learning vs. Deep Networks vs. Time-Series Foundation Models

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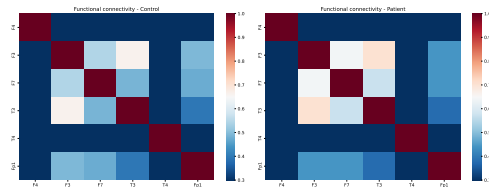
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- 2 Data and pipeline
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*One question: can we detect schizophrenia from EEG **without cheating**?*

Motivation: an objective biomarker for schizophrenia

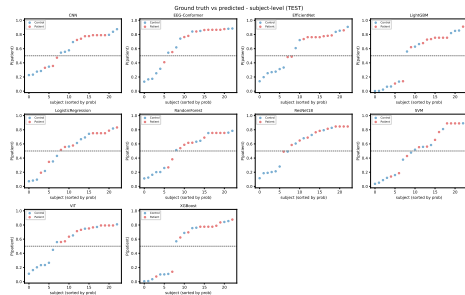
- Diagnosis is **clinical** $\Rightarrow$ inter-rater variability.
- EEG is **non-invasive**, **cheap**, high temporal resolution.
- Hypothesized markers: altered **connectivity** and **rhythmicity**.
- **Goal:** patient vs. control from resting EEG (ASZED, 153 subjects).



Healthy brains are highly synchronous; patients are not.

The methodological trap: data leakage

- Many windows per subject $\Rightarrow$ random split leaks the subject's **fingerprint**.
- The model memorizes the *person*, not the *pathology* $\Rightarrow$ inflated scores.
- **Our protocol:** strict **subject-wise** partition — all windows of a subject in one split only.
- Report **subject-level** metrics: a diagnosis concerns a patient, not a 5-second epoch.



Per-subject prediction vs. truth (controls low, patients high).

This single decision is what separates honest results from fiction.

ASZED — 153 subjects (76 patients / 77 controls)

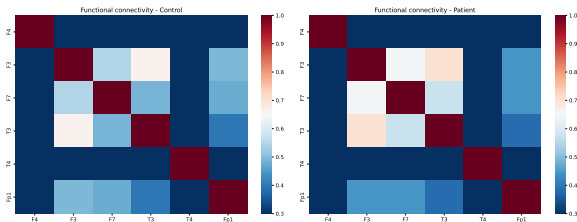
	Train	Val	Test	Total
Subjects	107	23	23	153
Windows	4,333	749	864	5,946

8-stage reproducible pipeline

- Preprocess: 0.5–45 Hz, 50 Hz notch, 250 Hz, 5 s windows, z-score
- EDA & channel selection
- tsfresh features (memory-safe) → SHAP funnel
- STFT/CWT/FFT images
- Classical ML / Deep nets / Chronos
- Window- & subject-level evaluation

Subject-wise split + per-channel .parquet cache $\Rightarrow$ honest *and* cheap to re-run.

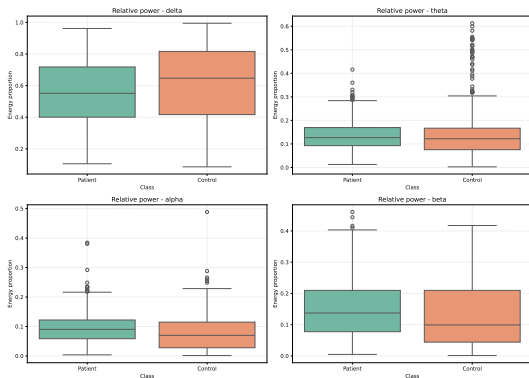
Clinical EDA (1): cortical disconnection



- Average inter-channel correlation.
- Controls: synchrony > 0.8 .
- Patients: drops to **0.3–0.7**.

Loss of long-range coherence — one of the most reproducible EEG findings in schizophrenia.

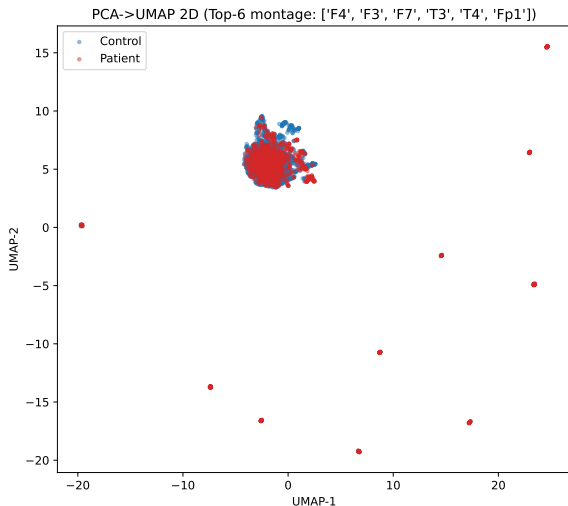
Clinical EDA (2): hyperfrontality & spectral shift



- Most discriminative region: **fronto-temporal**.
- Patients: *rightward spectral shift*
 - delta -0.066
 - beta $+0.018$, **gamma** $+0.031$
- Motivates the **Top-6** montage {F4,F3,F7,T3,T4,Fp1}.

Consistent with reported gamma-band dysregulation.

Channel selection and separability



- PCA→UMAP of windows (Top-6).
- **Partial** class separability $\Rightarrow$ moderately hard task.
- Six channels cut input size by $\sim 68\%$ vs. 18/19.

The clusters overlap — so the task is real, not a leakage artifact.

Tabular branch (tsfresh)

- 4,662 features (memory-safe, channel-by-channel + gc).
- Funnel: variance $\rightarrow$ corr $>$ 0.85 $\rightarrow$ **SHAP** $\Rightarrow$ 51 features.

4,662 $\rightarrow$ 1,800 $\rightarrow$ 1,204 $\rightarrow$ 51

Three model families

- **Classical:** LogReg, SVM, RF, XGBoost, LightGBM.
- **Deep (STFT):** CNN, ResNet18 (unfreeze layer4), EfficientNet, ViT.
- **Foundation:** Chronos — *frozen* encoder + linear head (zero-/few-shot).

Why STFT over wavelets? $\sim$ 1M CWT convolutions locally — and it does *not* help (shown later).

Main results — subject-level (Top-6)

Model	F1	ROC-AUC	Bal. Acc
Chronos (Top-18, winner)	0.846	0.826	0.833
ViT (STFT)	0.815	0.803	0.792
ResNet18 (transfer)	0.769	0.826	0.746
EfficientNet	0.720	0.735	0.701
EEG-Conformer	0.714	0.727	0.663
RF / XGB / LGBM / LogReg	0.692	0.67–0.73	0.659
CNN (from scratch)	0.615	0.652	0.572

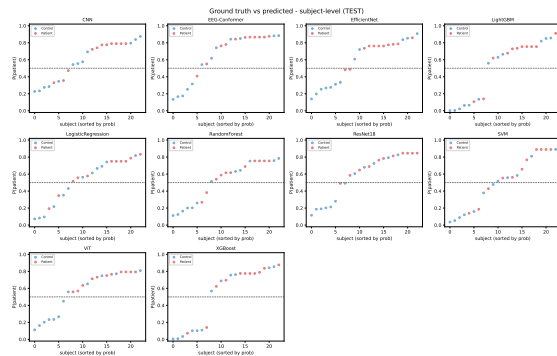
- Pre-trained & foundation models lead; classical models **tie** and **beat the from-scratch CNN**.
- Transfer of knowledge > raw capacity under limited clinical data.

The winner: a foundation model never trained on EEG

Chronos (frozen) + logistic head:

- **F1 = 0.846, Acc = 0.826, ROC-AUC = 0.826** (Top-18, subject-level).
- Best window-level **ROC-AUC = 0.859**.

It tokenizes a series into a “*language*” of trends, autocorrelation and regime changes — priors that transfer to EEG.



Ablation 1 — channels: Top-6 vs. Top-18

Model	Top-6		Top-18	
	F1	AUC	F1	AUC
ViT (STFT)	0.815	0.803	0.741	0.727
ResNet18	0.769	0.826	0.769	0.841
EfficientNet	0.720	0.735	0.769	0.780
CNN	0.615	0.652	0.750	0.841
EEG-Conformer	0.714	0.727	0.800	0.773
Chronos	—	0.859*	0.846	0.826

- Gains from 6→18 channels are **marginal**; Chronos AUC barely moves (0.859 → 0.856, *window-level).
- **Six fronto-temporal channels suffice** — 3× cheaper.

Ablation 2 — representation: do wavelets help?

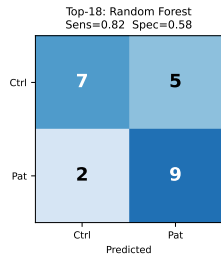
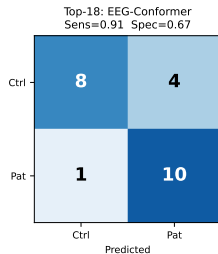
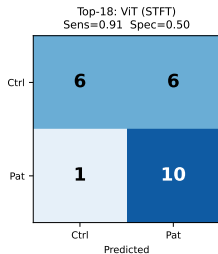
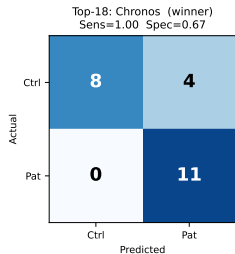
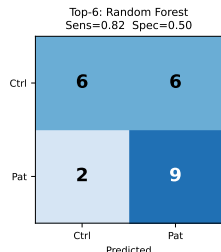
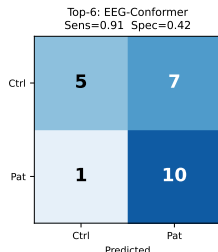
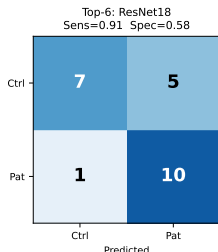
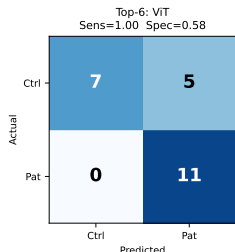
Subject-level F1 (Top-18)			
Arch.	STFT	CWT	FFT
CNN	0.750	0.640	0.741
ResNet18	0.769	0.692	0.786
ViT	0.741	0.786	0.815
EfficientNet	0.769	0.714	0.786

- **Wavelets (CWT) lose** — last in every row.
- STFT best at window level; **FFT** strongest at subject level.
- Caveat: CWT used 6 channels (vs. 18) — a bias *against* it.

Fixed-resolution STFT already suffices for band-limited resting rhythms.

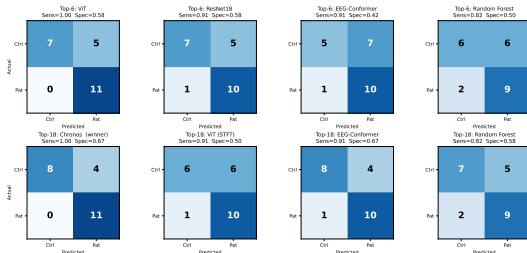
Error structure: confusion matrices (Top-6 vs. Top-18)

Subject-level confusion matrices (TEST, n=23: 12 controls, 11 patients): Top-6 (top) vs Top-18 (bottom)



What the confusion matrices tell us

Subject-level confusion matrices (TEST, n=23: 12 controls, 11 patients): Top-6 (top) vs Top-18 (bottom)

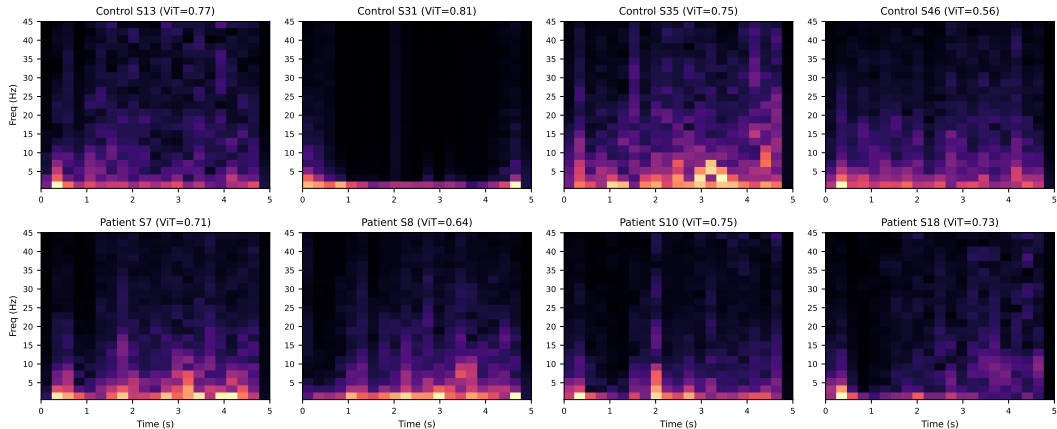


- **Asymmetric errors:** near-perfect **sensitivity** (0–3 patients missed); **specificity** is the bottleneck.
- **Chronos@18:** 0 false negatives, only 4 false positives — the best operating point.
- **6→18 channels** mainly cut false alarms (Conformer & CNN: TN 5 → 8), not missed patients.

*High-sensitivity profile = ideal for a **screening triage**: missing a patient costs more than a false alarm.*

Qualitative: what the models see

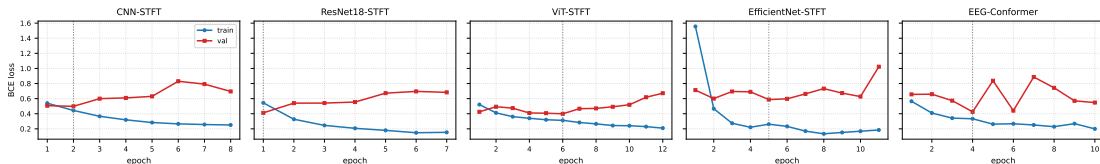
Qualitative STFT spectrograms (Top-6 mean) — controls vs patients, with mean predicted P(patient)



Patient spectrograms (bottom) show **more diffuse, higher-entropy** energy; ViT scores follow the visual cues, and **errors fall on the ambiguous cases**.

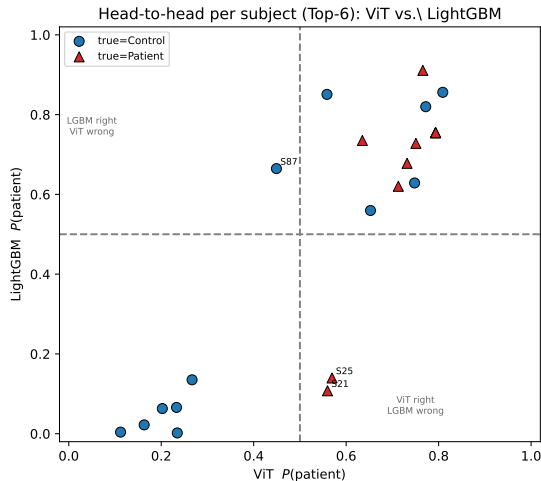
Qualitative: training dynamics expose overfitting

Training vs. validation loss (18-ch STFT run); dotted line = best-val epoch (early stopping)



From-scratch **CNN/EfficientNet** **overfit early** (val loss diverges → early stop); pre-trained **ViT/ResNet18** **stay stable** — the optimization-level signature of successful transfer.

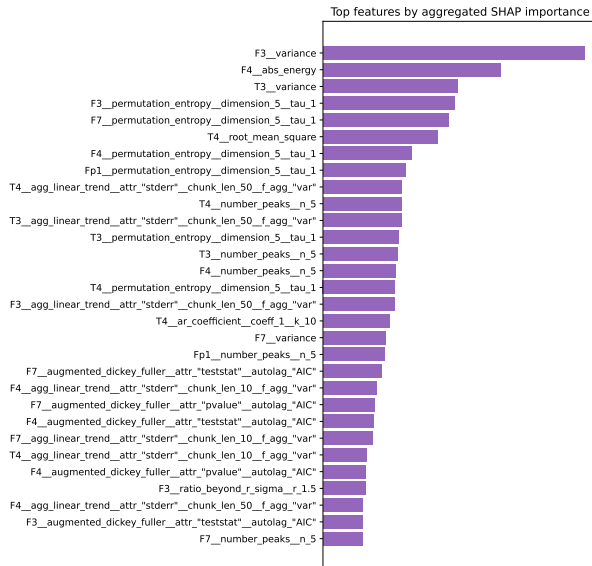
Qualitative: models are **complementary**



- Deep signal model (Conformer) vs. classical (SVM).
- Each is uniquely right on **2** subjects the other misses.
- Errors concentrate near the 0.5 threshold.

Partly decorrelated errors $\Rightarrow$ a stacked ensemble could beat any single model.

Explainability: SHAP → clinical reading



- Top features are **all fronto-temporal** (F3, F4, T3, T4, F7, Fp1).
- **Variance / energy** = cortical power (hyperfrontality).
- **Permutation entropy** = signal complexity / disorganization.
- **AR(1) & ADF stationarity** = altered temporal dynamics.

Statistics converge with known pathophysiology.

- **Method:** subject-wise partition + subject-level metrics are *indispensable* for honest EEG benchmarking.
- **Result:** a **frozen foundation model (Chronos)** wins — F1 0.846, ROC-AUC 0.826 — beating bespoke deep nets.
- **Localization:** six fronto-temporal channels suffice (hyperfrontality + cortical disconnection, confirmed by SHAP).
- **Representation:** wavelets do not beat STFT/FFT.
- **Trade-off:** interpretable classical pipelines stay competitive and auditable at minimal cost.

Future work: fine-tune Chronos on EEG; channel-matched CWT; stacked ensemble; external validation.

Thank you!

Questions?

Code & data:

`github.com/ricardoamiel/EEG-signal-prediction-for-schizophrenia-detection`

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